

Acoustic Differences in

North vs. South Peninsular Spanish

A formant-based vowel space analysis using Mozilla Common Voice

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Speech Signal Processing · Prof. Wolfgang Wokurek · 2026

Research Question & Motivation



Do speakers from Northern and Southern Peninsular Spain produce acoustically distinct vowel spaces?

Dialect Geography

Norte vs. Sur show known phonological differences, but acoustic evidence is limited

Acoustic Phonetics

F1/F2 formants directly encode vowel height and backness, the heart of dialect variation

Open Data

Mozilla Common Voice provides dialect-labelled recordings at scale, a rare resource for Spanish

Data & Preprocessing

Source

Mozilla Common Voice 25.0 (March 2026)
Spanish validated clips: dialect-labelled subset

Pipeline

1. **Filter by dialect label**
   España: Norte peninsular / Sur peninsular
2. **Random sample**
   300 clips per dialect → 600 total
3. **Copy audio files**
   preprocesador.py → audios_filtrados/
4. **Save metadata**
   metadata_filtrada.csv (path + dialecto)

600
clips sampled
(300 per dialect)

25.0
Common Voice
corpus version

Dialect Labels

Norte
España: Norte peninsular

Sur
España: Sur peninsular

Formant Extraction Method

1. **Load audio**
   parselmouth.Sound() —> each .mp3 file

2. **Find syllable nucleus**
   Peak intensity frame ⟶ time anchor

3. **Extract 50 ms window**
   ±25 ms around peak, clipped to audio bounds

4. **Sample 10 time points**
   np.linspace() across the window

5. **Burg LPC formant analysis**
   max_formants=5, ceiling=5000 Hz (mixed gender)

6. **Compute median F1 & F2**
   np.nanmedian() — ignores unvoiced frames

Tools: Python · parselmouth · Praat Burg algorithm · extractor_v2.py

Why these choices?

Intensity peak
Targets the vowel nucleus, most stable formant region

50 ms window
Captures steady-state, avoids consonant coarticulation

Median not mean
Robust to NaN from unvoiced frames (Praat returns NaN for silence)

5000 Hz ceiling
Safer for mixed-gender data than the default 5500 Hz

Cleaning & Quality Control



Clips sampled

600 · 300 Norte + 300 Sur

Formant extraction

903 · Including orphan files from prior run

Remove orphans

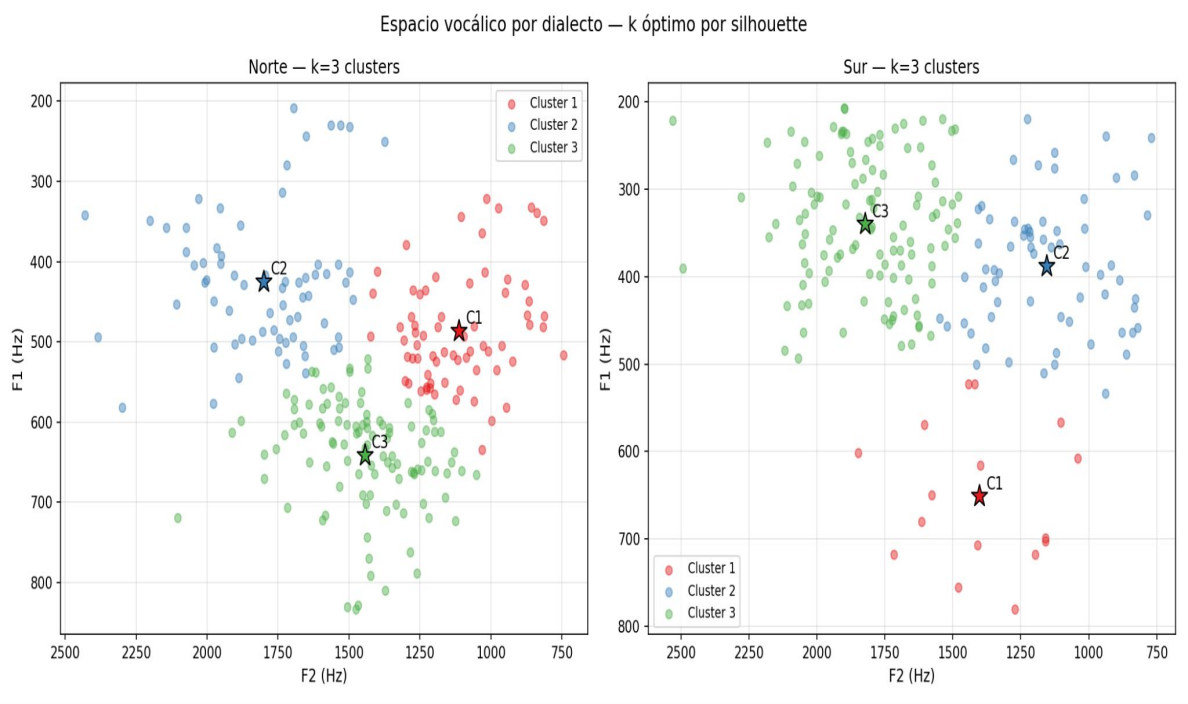
→ drop NaN dialect · Clips with no dialect label excluded

Acoustic filter

569 · F1: 200–900 Hz · F2: 700–2800 Hz

Final dataset: **244 Norte** + **196 Sur** = **440 clean tokens**

The Vowel Space: Norte vs. Sur



What we see

- Norte sits higher (larger F1 → more open vowels)
- Sur clusters lower-left (smaller F1, larger F2)
- Both dialects show k=3 optimal clusters
- Overlap exists — no perfect boundary

F1 axis inverted (phonetic convention: open vowels at top)

Statistical Results: F1 & F2

F1 — Vowel Height

536.7 Hz	380.7 Hz	d = 1.34
Norte mean F1	Sur mean F1	Cohen's d (large effect)

p < 0.001 · highly significant

F2 — Vowel Backness

1450 Hz	1567 Hz	d = −0.33
Norte mean F2	Sur mean F2	Cohen's d (small effect)

p < 0.001 · significant but smaller effect

Linguistic reading

Norte
Higher F1 → more open vowel production. Larger vowel space overall.

Sur
Lower F1 → closer/more reduced vowels. Slightly more front (higher F2).

F1 is the main discriminating feature between the two dialects.

Unexpected Finding: Bandwidth B2

B2 Norte = 152.5 Hz · **B2 Sur = 408.4 Hz** · *Cohen's d = −1.39 (largest effect in the study)*

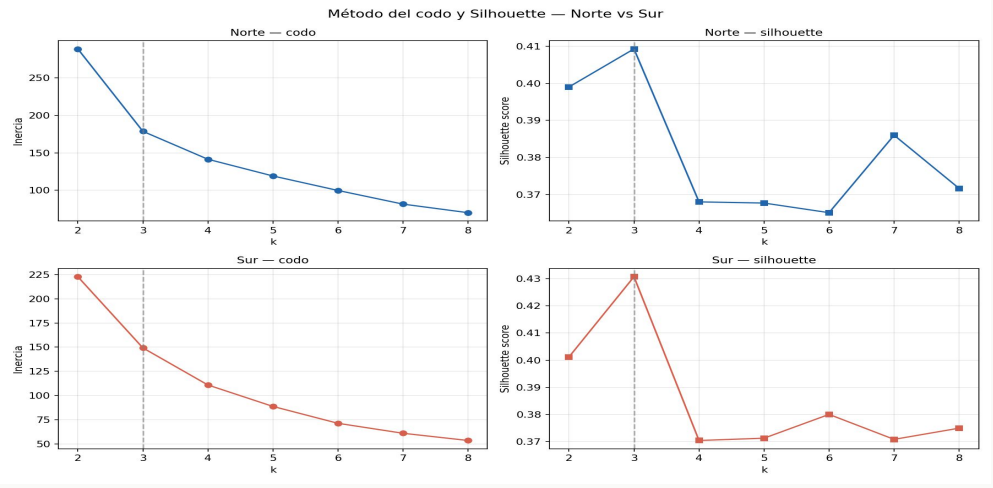
What does B2 measure?

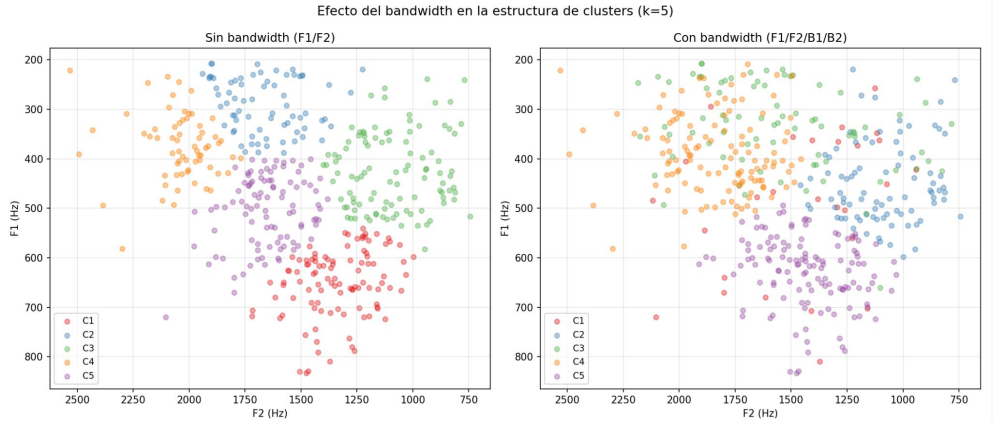
Bandwidth is the width of the formant resonance peak. A wider B2 indicates a less distinct / more damped F2 resonance, associated with greater vocal tract damping or more variable articulation.

Possible interpretations

Sur speakers may produce more reduced or centralised vowels, or the dialect involves more coarticulation. This requires further investigation with phoneme-level labels, which were not available in this study.

Clustering Analysis (K-Means)





Optimal k = 3 (both dialects)

Elbow method + silhouette agree: 3 clusters describe the F1/F2 space best — likely corresponding to vowel categories

Adding bandwidth lowers silhouette

Silhouette drops from 0.370 → 0.298 when B1/B2 are included. Bandwidth captures different structure — not useful for vowel space clustering

Sur internal structure: k=5

Sur shows finer subgrouping (k=5) but low silhouette (0.26–0.28). Without province metadata, these subgroups cannot be interpreted

Limitations & Methodological Notes

No vowel-level labels

Analysis covers the global vowel space, not individual phonemes. Vowel labelling was abandoned due to circularity: acoustic ranges needed to assign labels, labels needed to define ranges.

Self-reported dialect labels

Mozilla Common Voice labels come from speakers themselves. Actual regional provenance is not verified, some mislabelling is likely.

No phoneme-level forced alignment

Intensity peak targets the syllable nucleus, not a specific phoneme. Some tokens may contain non-vowel noise.

Sample imbalance after cleaning

244 Norte vs. 196 Sur after filtering. Difference is modest but could bias distributional comparisons.

Future Work & Extensions

Add Mexican Spanish

Three-way comparison (Norte / Sur / Mexico) would test whether differences are Iberian-specific or broader

Forced alignment + labelling

Use WebMAUS or MFA to get phoneme-level segments, enables vowel-by-vowel F1/F2 comparison

Province-level metadata

Finer dialect granularity could explain the Sur k=5 substructure and the B2 difference

Classifier as evaluation

Train a simple LDA or SVM on F1/F2/B2, dialect separability score as an objective measure

Conclusions

01 F1 is significantly higher in Norte: consistent with a more open vowel space in Northern Peninsular Spanish

02 F2 is also significantly different but effect is small: Sur is slightly more front, but the distinction is secondary

03 B2 is the biggest effect (d = 1.39): Sur shows markedly wider F2 bandwidth, suggesting more variable or reduced articulation

04 K-means finds 3 clusters in both dialects: structure consistent with Spanish vowel categories /a, e-i, o-u/

05 Bandwidth adds different information: not useful for vowel space clustering, but phonetically meaningful on its own



Speech
Signal
Processing

Prof. Wolfgang Wokurek
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Thank you!

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Questions?

· *Mozilla Common Voice 25.0 (March 2026)*

· *Parselmouth / Praat —> Burg LPC formant analysis*

· *Cohen's d —> effect size for dialect comparison*